

Data Structures

Lecture 4

Stack

Abstract Data Type

References:

Data Structures and Algorithms in Java

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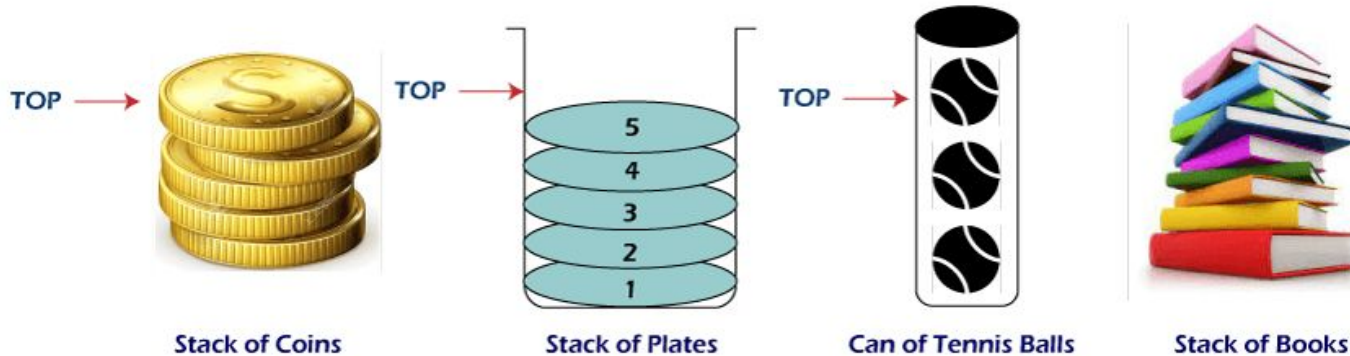
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Topics

- Stack Abstract Data Type
 - Introduction to Stack ADT
 - Stack Operations



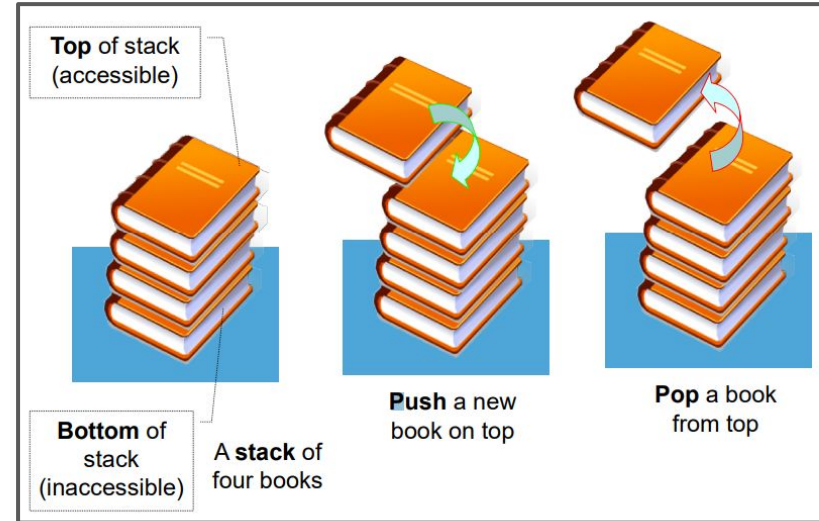
Stack ADT

Introduction to Stack ADT

- Operations are done at the first (top) position only.
- The latest item added is the first item can get out from the stack:

Known as ***Last In First Out*** (LIFO).

- Real life example: A stack of books.
 - It is possible to see the top book only
 - To access any other book, we need to remove the books on top of it.

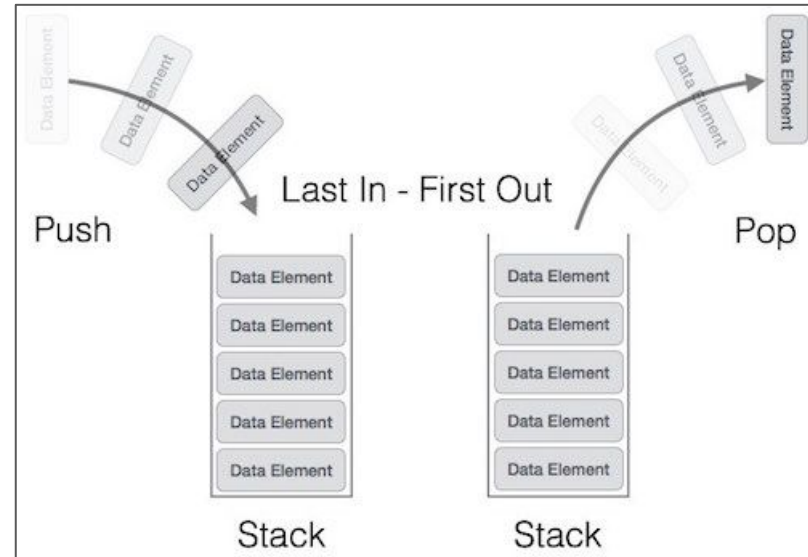


Stack ADT

Introduction to Stack ADT

Stack applications:

- Parsing code
- Dealing with function calls
- Tracking undo and redo operations
- Matching parentheses in C++
- Going forward and back in a web browser



Stack ADT

Introduction to Stack ADT

Push - Add the new element to the stack

- The top pointer points to the newest added element.
- **Stack Overflow problem:** when trying to push items on stack greater than the size of the stack

push_method

STEP 1 START

STEP 2 Store the element to push into "num"

STEP 3 If the stack is full

show error ("stack is full ")

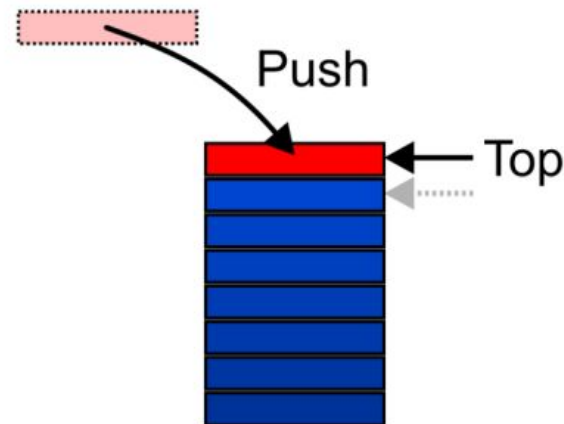
exit

else

Increment top: $top = top + 1$

$stack[top] = num$

STEP 4 STOP



Stack ADT

Introduction to Stack ADT

For the following stack, the top points at 12

12
41

<- Top of the Stack

To Push 37 into the stack, we need first to increase the top value

12
41

<- Top of the Stack

Then add the value 37 at the top pointed element

37
12
41

<- Top of the Stack

Stack ADT

Introduction to Stack ADT

Pop - Remove the last element on the stack and returns it

- The top pointer points to the element below the top element.
- **Stack Underflow problem:** when trying to pop an element from an empty stack.

pop_method

STEP 1 START

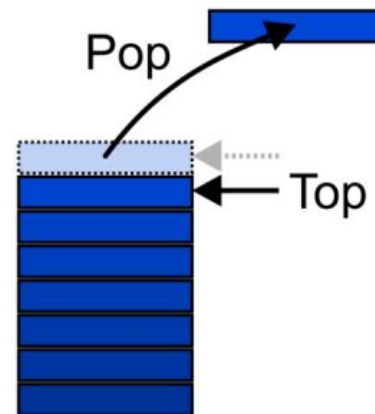
STEP 2 If the stack is empty
 produce an error ("Stack is empty")
 exit.

else

 return stack[top]

 Decrease the value of top by 1.

STEP 3 STOP



Stack ADT

Introduction to Stack ADT

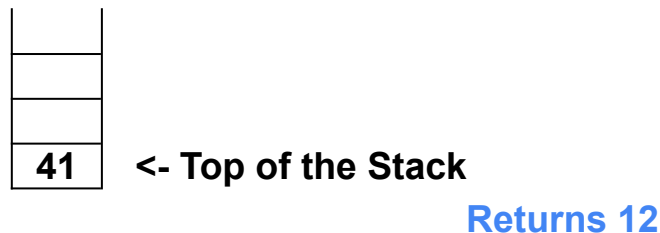
For the following stack, the top points at 37



To pop the element 12, we need first to pop the element 37



Then add the value 37 at the top pointed element



Stack ADT

Introduction to Stack ADT

Peek - Returns the top element from the stack

No addition or deletion is involved in this operation

peek_method

STEP 1 START

STEP 2 return stack[top]

STEP 3 STOP

Let's consider the stack below , the top points at 37

37
12
41

<- Top of the Stack

The Peek **returns 37**, no changes to the stack are made.



Stack ADT

Introduction to Stack ADT

isFull – check if the stack is full.

No addition or deletion is involved in this operation

isFull_method

STEP 1 START

STEP 2 If the stack is full

Return TRUE

else

Return FALSE

STEP 4 STOP

41

<- Top of the Stack

Returns FALSE

10
37
12
41

<- Top of the Stack

Returns TRUE



Stack ADT

Introduction to Stack ADT

isEmpty - Returns if the stack is empty or not

No addition or deletion is involved in this operation

isEmpty_method

STEP 1 START

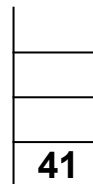
STEP 2 If the stack is empty

Return TRUE

else

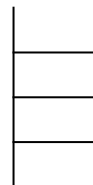
Return FALSE

STEP 4 STOP



<- Top of the Stack

Returns FALSE



<- Top of the Stack

Returns TRUE

Stack ADT

Introduction to Stack ADT

Types of Stack

Register Stack

- The stack depth is finite, it handles only a small amount of data

Memory Stack

- The stack depth is flexible, it handles a large amount of data.

Stack Applications

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